AI Career Pro

Full-Stack AI-Powered Career Platform — Project Explanation Document

1. Project Overview

AI Career Pro is a full-stack web application that helps job seekers with:

- **AI Resume Builder** — Build resumes with AI-generated bullet points and download as PDF
- **Resume Analyzer** — Upload a resume (paste text or PDF) and get ATS score, skill gaps, and suggestions
- **Mock Interview Trainer** — Practice interviews with AI-generated questions, answer via text or voice
- **Career Guidance** — Get personalized career roadmaps, skill recommendations, and project ideas
- **Dashboard** — Track all progress with charts and analytics

2. Technology Stack

Layer	Technology	Purpose
Frontend	React 19 + Vite 8	Fast, modern SPA framework
Styling	Tailwind CSS 4	Utility-first CSS styling
UI Components	Radix UI + ShadCN	Accessible, reusable UI components
Animations	Framer Motion	Smooth page transitions and micro-animations
Charts	Recharts	Dashboard data visualization
Backend	Node.js + Express	REST API server for AI features
AI Engine	Google Gemini 2.0 Flash	Powers all AI features (analysis, interviews, guidance)
Database	Supabase (PostgreSQL)	User data, resumes, sessions storage
Authentication	Supabase Auth	Email/password sign-up and login
PDF Parsing	pdf-parse-fork	Extract text from uploaded PDF resumes
File Upload	Multer	Handle file uploads (PDF, audio)
Notifications	Sonner	Toast notifications for user actions

3. Architecture Diagram

```
┌──────────────────────────────────────────────────────────┐
│                    FRONTEND (React + Vite)               │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐    │
│  │ Landing  │ │Dashboard │ │ Resume   │ │ Interview │    │
```

```
│   │  Page     │ │  Page     │ │ Builder   │ │   Page      │  │
│   └──────────┘ └──────────┘ └──────────┘ └──────────────┘  │
│                                                              │
│   ┌──────────┐ ┌──────────┐ ┌──────────────────────────┐  │
│   │ Resume   │ │ Career   │ │  Auth Context + Supabase │  │
│   │ Analyzer │ │ Guidance │ │     Client Library       │  │
│   └──────────┘ └──────────┘ └──────────────────────────┘  │
└──────────────────────────────────────────────────────────┘
                        │
                        │ HTTP (via Vite proxy /api → :3001)
┌──────────────────────▼──────────────────────────────────┐
│              BACKEND (Express.js — Port 3001)            │
│   ┌──────────────────────────────────────────────┐      │
│   │  API Routes: /api/ai/analyze-resume          │      │
│   │              /api/ai/generate-bullets         │      │
│   │              /api/ai/generate-questions       │      │
│   │              /api/ai/evaluate-answer          │      │
│   │              /api/ai/career-guidance          │      │
│   │              /api/ai/parse-pdf                │      │
│   │              /api/ai/transcribe               │      │
│   └──────────────────────────────────────────────┘      │
│                         │                                │
│                         │                                │
│   ┌──────────────────────▼──────────────────────┐      │
│   │          Gemini AI (gemini.js)               │      │
│   │   geminiChat() → text generation             │      │
│   │   geminiTranscribe() → audio-to-text         │      │
│   └──────────────────────────────────────────────┘      │
└──────────────────────────────────────────────────────────┘
                        │
┌──────────────────────▼──────────────────────────────────┐
│                SUPABASE (Cloud Database)                 │
│   profiles │ resumes │ resume_analyses                   │
│   interview_sessions │ career_roadmaps                   │
│   + Row Level Security (RLS) + Auth                      │
└──────────────────────────────────────────────────────────┘
```

4. Folder Structure

```
manali-project1/
├── public/                    # Static assets
├── src/                       # Frontend source code
│   ├── App.jsx                # Main app — routing setup
│   ├── main.jsx               # Entry point — renders App
│   ├── index.css              # Global styles (Tailwind)
│   ├── context/
│   │   └── AuthContext.jsx    # Authentication state management
│   ├── lib/
│   │   ├── supabase.js        # Supabase client initialization
│   │   ├── database.js        # Database helper functions (CRUD)
│   │   └── utils.js           # Utility functions (cn, etc.)
│   ├── components/
│   │   ├── layout/
│   │   │   ├── AppShell.jsx   # Main app layout with sidebar
│   │   │   ├── Sidebar.jsx    # Navigation sidebar
│   │   │   └── ProtectedRoute.jsx  # Auth guard for routes
│   │   └── ui/                # Reusable UI components
```

```
│   │       ├── button.jsx, card.jsx, input.jsx, ...
│   └── pages/
│       ├── LandingPage.jsx    # Public homepage
│       ├── LoginPage.jsx      # Login form
│       ├── SignupPage.jsx     # Registration form
│       ├── DashboardPage.jsx  # User dashboard with charts
│       ├── ResumeBuilderPage.jsx   # AI resume builder
│       ├── ResumeAnalyzerPage.jsx  # Resume analysis tool
│       ├── InterviewPage.jsx       # Mock interview trainer
│       ├── CareerGuidancePage.jsx  # Career roadmap generator
│       └── ProfilePage.jsx        # User profile settings
├── server/                    # Backend
│   ├── index.js               # Express server + API routes
│   ├── gemini.js              # Google Gemini AI integration
│   └── package.json           # Backend dependencies
├── supabase/
│   └── schema.sql             # Database schema (5 tables + RLS)
├── .env                       # Environment variables
├── vite.config.js             # Vite configuration + proxy
└── package.json               # Frontend dependencies
```

5. Explanation of Each Part

5.1 — App.jsx (Main Router)

This is the entry point of the app. It sets up:

- **Public routes:** Landing page (`/`), Login (`/login`), Signup (`/signup`)
- **Protected routes:** Dashboard, Resume Builder, Resume Analyzer, Interview, Career Guidance, Profile — all wrapped in `ProtectedRoute` so only logged-in users can access them
- **AuthProvider:** Wraps the entire app to provide authentication state everywhere
- **Toaster:** Global toast notification system using Sonner

5.2 — AuthContext.jsx (Authentication)

Manages user authentication with two modes:

- **Live Mode:** Uses Supabase Auth for real sign-up/sign-in with email and password
- **Demo Mode:** If Supabase isn't configured, auto-logs in with a demo user so the app still works
- Provides `signUp()` , `signIn()` , `signOut()` functions to all components via React Context
- Listens for auth state changes in real-time using Supabase's `onAuthStateChange`

5.3 — Supabase Client (supabase.js)

Initializes the Supabase client using environment variables (`VITE_SUPABASE_URL` and `VITE_SUPABASE_ANON_KEY`). This client is used for both authentication and database operations.

5.4 — Database Layer (database.js)

Contains all CRUD helper functions organized by feature:

Function Group	Functions	Purpose
Profiles	`getProfile()`, `updateProfile()`	Read/update user profile data
Resumes	`getResumes()`, `saveResume()`, `deleteResume()`	CRUD for user resumes
Analyses	`saveAnalysis()`, `getAnalyses()`	Store and retrieve resume analysis results
Interviews	`saveInterviewSession()`, `getInterviewSessions()`	Save mock interview sessions
Roadmaps	`saveRoadmap()`, `getRoadmaps()`	Save career guidance results
Dashboard	`getDashboardStats()`	Aggregate stats for dashboard display

5.5 — Backend Server (server/index.js)

An Express.js server running on **port 3001** that provides 7 API endpoints:

Endpoint	Method	What It Does
`/api/ai/parse-pdf`	POST	Extracts text from uploaded PDF files
`/api/ai/analyze-resume`	POST	Sends resume text to Gemini AI for ATS scoring and analysis
`/api/ai/generate-bullets`	POST	Generates 4 professional bullet points for a given job role
`/api/ai/generate-questions`	POST	Generates 5 interview questions for a given role
`/api/ai/evaluate-answer`	POST	Evaluates an interview answer with scores and feedback
`/api/ai/career-guidance`	POST	Creates a personalized career roadmap with skills and projects
`/api/ai/transcribe`	POST	Converts voice audio to text using Gemini's multimodal capability

5.6 — Gemini AI Integration (gemini.js)

Handles all AI communication with Google Gemini 2.0 Flash model:

- **`geminiChat(systemPrompt, userMessage)`** — Sends a system prompt + user message, returns clean JSON response (strips markdown code fences)
- **`geminiTranscribe(buffer, filename)`** — Sends audio buffer (base64) to Gemini for speech-to-text transcription

5.7 — Pages Overview

Page	Route	Description
LandingPage	`/`	Public homepage with hero section, features showcase, stats, and CTA. Uses Framer Motion animations.
LoginPage	`/login`	Email + password login form with validation. Redirects to dashboard on success.
SignupPage	`/signup`	Registration form with full name, email, password. Creates profile automatically.
DashboardPage	`/dashboard`	Displays score cards (Resume, Interview, Skills, ATS), interview progress chart, skills overview, recent activity, and improvement tips.
ResumeBuilderPage	`/resume-builder`	Multi-section form to build resumes. AI generates bullet points. Supports multiple templates. Export as PDF.
ResumeAnalyzerPage	`/resume-analyzer`	Paste resume text or upload PDF. AI returns ATS score, strengths, weaknesses, skill gaps, and section-wise feedback.
InterviewPage	`/interview`	Select a role → AI generates 5 questions → Answer via text or voice recording → AI evaluates each answer with detailed scores.
CareerGuidancePage	`/career-guidance`	Enter interests/goals → AI generates a full career roadmap with skills to learn, priority levels, resources, and project ideas.
ProfilePage	`/profile`	View and edit personal details: name, phone, location, LinkedIn URL, avatar.

5.8 — Database Schema (schema.sql)

5 PostgreSQL tables with Row Level Security (RLS):

Table	Key Columns	Purpose
`profiles`	full_name, avatar_url, phone, location, linkedin	Extends Supabase Auth users with profile data
`resumes`	title, template, data (JSONB)	Stores resume content and template choice
`resume_analyses`	resume_text, ats_score, analysis (JSONB)	Stores AI analysis results
`interview_sessions`	role, questions, answers, evaluations (JSONB), average_score	Stores complete interview sessions

`career_roadmaps`	interests, roadmap (JSONB)	Stores generated career roadmaps

Security: RLS policies ensure each user can only read/write their own data. A database trigger auto-creates a profile when a new user signs up.

5.9 — UI Components

Reusable ShadCN/Radix-based components:

- `button.jsx` — Styled button with variants (default, outline, ghost, etc.)
- `card.jsx` — Card container with header, content, title, description
- `input.jsx` / `textarea.jsx` / `label.jsx` — Form input elements
- `tabs.jsx` — Tab navigation component
- `badge.jsx` — Status badges (success, warning, etc.)
- `progress.jsx` — Progress bar for skills visualization

5.10 — Layout Components

- **AppShell.jsx** — Main layout wrapper with sidebar navigation and content outlet
- **Sidebar.jsx** — Left navigation sidebar with links to all features and user info
- **ProtectedRoute.jsx** — Route guard that redirects to login if user is not authenticated

5.11 — Configuration Files

- **vite.config.js** — Configures Vite with React plugin, Tailwind CSS plugin, path alias (`@` → `src/`), and proxy (`/api` → `localhost:3001`)
- **.env** — Stores environment variables: `VITE_SUPABASE_URL` , `VITE_SUPABASE_ANON_KEY` , `GEMINI_API_KEY`

6. How to Run the Project

1. Install frontend dependencies: `npm install` (in root folder)
2. Install backend dependencies: `cd server && npm install`
3. Set up `.env` file with your Supabase URL, Anon Key, and Gemini API Key
4. Run the Supabase schema: paste `supabase/schema.sql` in Supabase SQL Editor
5. Start backend: `cd server && npm run dev` (runs on port 3001)
6. Start frontend: `npm run dev` (runs on port 5173)
7. Open `http://localhost:5173` in your browser

7. User Flow Summary

```
User visits Landing Page → Signs Up / Logs In
        ↓
   Dashboard (see progress charts, scores, tips)
        ↓
   ┌── Resume Builder ──→ Create resume → AI bullets → Download PDF
   ├── Resume Analyzer ──→ Upload/paste resume → AI ATS score + feedback
```

```
    ├── Mock Interview ──→ Select role → AI questions → Answer → AI evaluation
    └── Career Guidance ─→ Enter goals → AI roadmap with skills + projects
         ↓
    Profile → Edit personal details
```